TELECOM CUSTOMER CHURN ANALYSIS AND PREDICTION

Machine Learning Project Report

Project	Customer Churn Analysis and Prediction
Dataset	IBM Telco Customer Churn
Primary model	Gradient Boosting Classifier
Evaluation approach	80/20 stratified train-test split
Date	November 2026

Prepared as a technical report documenting the data preparation, feature selection, model comparison, model training, and evaluation stages of a telecom customer churn prediction project.

Executive Summary

Customer churn occurs when a customer stops using a company’s service. For telecommunications providers, identifying customers who may leave can help prioritize retention activities and improve customer experience. This project uses the IBM Telco Customer Churn dataset to build and evaluate a supervised machine learning model that predicts whether a customer will churn.

The workflow covered all six project tasks: data preparation, data splitting, feature selection, model selection, model training, and model evaluation. A Gradient Boosting Classifier was selected after five-fold stratified cross-validation. On the held-out test set, it achieved 80.27% accuracy and an ROC-AUC of 84.19%. Its churn-class recall was 53.48%, indicating that further work is needed to identify a larger share of customers who actually leave.

1. Introduction

Customer retention is important in the telecommunications industry because losing existing customers can reduce recurring revenue and increase the cost of acquiring replacement customers. A churn prediction model can help a business identify customers who may be at risk and support more targeted retention decisions.

Project objective: develop a machine learning classification pipeline that predicts whether a telecom customer will churn (Yes) or remain with the service (No), and evaluate its performance using appropriate classification metrics.

2. Dataset Description

The project uses the IBM Telco Customer Churn dataset. The original dataset contains 7,043 customer records and 21 columns, including a customer identifier, demographic information, subscribed services, account information, charges, and the Churn target.

The customerID column was excluded from model inputs because it is an identifier rather than a meaningful customer characteristic. After removing this identifier, the modelling dataset contained 19 input features and one target column.

Target variable: Churn. The values were mapped to numeric labels for model training and evaluation: No = 0 and Yes = 1.

The target distribution was imbalanced: 5,174 customers (73.46%) were labelled No and 1,869 customers (26.54%) were labelled Yes. This imbalance makes it important to examine class-specific recall and F1-score rather than relying on accuracy alone.

3. Data Preparation

The data preparation process included checking the dataset structure, identifying missing values, checking duplicates, cleaning whitespace, and converting the TotalCharges column to a numeric data type.

- Whitespace was stripped from object/string columns to standardize text values.
- TotalCharges was converted to numeric using coercion for values that could not be parsed.
- This conversion revealed 11 missing TotalCharges values. All 11 belonged to customers with tenure equal to zero.
- The 11 TotalCharges values were filled with 0.0, consistent with zero tenure and no accumulated charges.
- No duplicate customer records were found in the dataset.
- customerID was removed from the model input features.

Categorical variables were encoded using OneHotEncoder with handle_unknown='ignore'. The preprocessing transformer was fitted within the modelling pipeline so that encoding was learned from training data rather than from the full dataset.

4. Train-Test Split

The data was split into training and testing subsets using an 80/20 split. Stratification was applied to preserve approximately the same churn-class proportions in both subsets. The random_state was set to 42 to make the split reproducible.

Subset	Records	Share
Training set	5,634	80%
Testing set	1,409	20%
Total	7,043	100%

The resulting training input had 5,634 rows and 19 columns; the test input had 1,409 rows and 19 columns. The target class proportions were maintained by stratification.

5. Feature Selection

Feature selection was performed to reduce the number of encoded input features while retaining useful information for churn prediction. Mutual information was used to estimate the relationship between each transformed feature and the target variable.

One-hot encoding expanded the 19 raw input columns into 45 transformed features. Different values of k were compared using five-fold stratified cross-validation and ROC-AUC. The tested values were 10, 15, 20, 25, 30, and 45.

Selected features (k)	Mean CV ROC-AUC
10	0.841325
15	0.842187
20	0.844431
25	0.844290
30	0.845611
45	0.846310

Thirty features were selected as a practical compromise between compactness and cross-validation performance. The mean ROC-AUC at k=30 was 0.845611, compared with 0.846310 when all 45 transformed features were used. The difference was small, so the 30-feature solution was retained to reduce model input dimensionality.

The selected features included contract type, tenure, monthly and total charges, internet service, online security, technical support, online backup, device protection, streaming services, payment method, and selected demographic/account attributes.

6. Model Selection

Five classification algorithms were compared using five-fold StratifiedKFold cross-validation. The primary comparison metric was ROC-AUC. Preprocessing and feature selection were placed inside each pipeline so they were fitted separately within each training fold.

Model	Mean CV ROC-AUC	Std. deviation
Gradient Boosting	0.846644	0.012677
Logistic Regression	0.845614	0.011783
Random Forest	0.818576	0.012857
K-Nearest Neighbors	0.777024	0.011818

Decision Tree	0.664930	0.009299

Gradient Boosting achieved the highest mean cross-validation ROC-AUC (0.846644) and was selected as the primary model. Logistic Regression was close at 0.845614, so the difference should not be interpreted as evidence of a large performance advantage.

7. Model Training

The final Gradient Boosting pipeline was trained using the full training subset. The pipeline included a ColumnTransformer for categorical encoding and numerical features, SelectKBest using mutual information with k=30, and a GradientBoostingClassifier.

The final classifier configuration was:

- n_estimators = 100
- learning_rate = 0.1
- max_depth = 3
- random_state = 42

The fitted pipeline was saved as customer_churn_model.joblib. Because preprocessing and feature selection are included in the saved pipeline, new customer records should be supplied in the original 19-feature format and processed by the pipeline.

8. Model Evaluation

The final model was evaluated once on the held-out test subset of 1,409 customers. The test set was not used to fit the model. The evaluation included accuracy, precision, recall, F1-score, ROC-AUC, a classification report, and a confusion matrix.

Metric	Test result
Accuracy	0.802697 (80.27%)
Precision (churn class)	0.657895 (65.79%)
Recall (churn class)	0.534759 (53.48%)
F1-score (churn class)	0.589971 (59.00%)
ROC-AUC	0.841879 (84.19%)

8.1 Confusion Matrix

The confusion matrix below compares actual customer outcomes with model predictions. Rows represent actual classes and columns represent predicted classes.

	Predicted: No Churn	Predicted: Churn
Actual: No Churn	931 (True Negative)	104 (False Positive)
Actual: Churn	174 (False Negative)	200 (True Positive)

- **True negatives (931):** customers who did not churn and were correctly predicted as No Churn.
- **False positives (104):** customers who did not churn but were predicted as Churn.
- **False negatives (174):** customers who churned but were predicted as No Churn.
- **True positives (200):** customers who churned and were correctly predicted as Churn.

8.2 Interpretation

The model correctly classified 80.27% of all test customers. Its ROC-AUC of 0.841879 indicates useful ability to distinguish between churn and non-churn customers across classification thresholds. However, churn recall was 53.48%, meaning that the model identified 200 of the 374 actual churners and missed 174. This is a material limitation for a retention use case, where failing to identify an at-risk customer may mean losing an opportunity to intervene.

Churn precision was 65.79%, meaning that about two-thirds of customers flagged as churners actually churned in this test set. The model therefore provides a useful starting point, but should not be treated as a fully optimized operational retention system.

9. Limitations

- The target classes are imbalanced, with fewer churn cases than non-churn cases.
- The churn-class recall of 53.48% means that a substantial number of actual churners were missed.
- The project uses a single train-test split for the final holdout evaluation; results may vary with another split or future customer data.
- The model is trained on the fields available in the selected dataset and may not capture all real-world reasons for churn, such as competitor offers, service incidents, or changes in customer circumstances.
- The model predicts association with churn, not the cause of churn. Predictions should not be interpreted as causal explanations.

10. Recommendations and Future Improvements

- Investigate probability-threshold tuning using training/validation data to increase churn recall while measuring the resulting precision trade-off. Do not tune the threshold on the final test set.
- Compare class-weighted models and resampling methods. Any resampling must occur inside cross-validation folds to avoid data leakage.
- Perform systematic hyperparameter tuning using cross-validation and retain a separate untouched test set for final evaluation.
- Evaluate business-oriented measures such as precision-recall AUC, expected retention campaign cost, and the value of correctly identifying at-risk customers.
- Monitor model performance over time and retrain when customer behaviour or service conditions change.
- Review feature importance and prediction explanations with domain experts before using model outputs to make customer-facing decisions.

11. Conclusion

This project developed a complete machine learning workflow for telecom customer churn analysis and prediction using the IBM Telco Customer Churn dataset. The work covered data cleaning, an 80/20 stratified split, mutual-information feature selection, cross-validation-based model comparison, final model training, and holdout evaluation.

Gradient Boosting was selected based on its cross-validation ROC-AUC and achieved 80.27% accuracy and 84.19% ROC-AUC on the test set. The results show that the model can distinguish churn and non-churn customers reasonably well, but its 53.48% churn recall highlights the need for further improvement before operational deployment. Overall, the project demonstrates how machine learning can support customer retention analysis while also showing why class-specific evaluation and business context matter.

12. Tools and Technologies

Tool / Library	Purpose

Python	Programming language
Pandas and NumPy	Data cleaning and numerical operations
Scikit-learn	Preprocessing, feature selection, model training, and evaluation
Matplotlib / Seaborn	Data visualization
Joblib	Saving the fitted model pipeline
Google Colab / Jupyter	Notebook-based development and experimentation
GitHub	Version control and project publication

References

- IBM Telco Customer Churn dataset, commonly distributed as the Telco Customer Churn sample dataset.
- Scikit-learn documentation: preprocessing, model selection, feature selection, ensemble methods, and classification metrics — https://scikit-learn.org/stable/

Reproducibility note: *The reported scores and dataset counts are based on the outputs recorded during this project workflow. Before formal submission, verify the final notebook, saved model, and CSV result files are included in the GitHub repository and that the notebook can be rerun from top to bottom.*